

ESSAOUIRA, Morocco

WMO: 602200

Lat: 31.5096N

Lon: 9.7726W

Elev: 26

StdP: 14.68

Time Zone: 1.00 (E01)

Period: 90-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 46.7	(c) 49.6	(d) 35.5	(e) 30.5	(f) 61.0	(g) 38.9	(h) 34.9	(i) 57.2	(j) 30.3	(k) 62.3	(l) 26.2	(m) 60.4	(n) 5.0	(o) 70	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.0	(c) 91.3	(d) 70.3	(e) 85.9	(f) 68.9	(g) 81.0	(h) 66.9	(i) 72.4	(j) 84.1	(k) 70.9	(l) 79.0	(m) 69.7	(n) 75.4	(o) 13.1	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	110.0	73.3	68.8	105.8	72.1	67.9	102.6	71.2	35.9	84.5	34.7	78.8	33.7	75.7	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 26.5	(b) 23.1	(c) 20.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	59.2	59.7	61.7	62.8	65.7	68.6	69.9	70.0	69.6	67.3	64.1	60.8
	DBStd	5.60	3.56	3.56	4.36	3.60	3.66	3.87	5.15	4.72	3.90	3.77	4.04	3.58
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	792	182	152	124	84	28	2	3	0	2	14	63	138
	CDD50	5465	287	272	361	385	486	558	615	620	587	537	423	334
	CDD65	783	3	4	21	19	49	110	154	155	139	86	36	7
	CDH74	3625	19	12	109	90	213	471	971	879	520	220	115	6
	CDH80	1429	1	0	23	31	73	185	472	412	163	54	15	0
(14) Wind	WSAvg	10.3	8.3	10.6	10.2	11.5	11.2	11.9	12.3	11.8	10.2	8.7	8.7	8.7
(15) Precipitation	PrecAvg	11.9	1.9	1.5	1.3	1.2	0.4	0.1	0.0	0.0	0.2	1.1	1.9	2.0
	PrecMax	18.0	6.9	6.0	4.9	4.2	1.4	0.6	0.2	1.0	1.1	5.0	8.0	6.6
	PrecMin	4.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.8	1.8	1.4	1.1	1.1	0.4	0.2	0.0	0.2	0.3	1.4	2.0	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	75.4	82.6	82.1	89.8	93.8	98.4	98.7	90.0	87.6	82.3	74.8
		MCWB	58.5	57.7	62.3	64.5	67.6	75.2	72.1	72.7	69.4	64.7	63.3	60.1
	2%	DB	71.7	71.5	76.7	74.8	79.1	83.9	90.6	89.3	84.1	79.1	77.3	70.6
		MCWB	59.5	60.7	60.5	61.9	65.2	68.5	71.3	70.8	68.9	65.5	63.5	62.8
	5%	DB	68.5	68.3	71.9	70.1	73.6	78.4	84.2	82.6	79.8	75.2	73.3	68.5
		MCWB	60.9	60.9	61.0	62.8	63.9	67.1	69.7	69.5	66.9	66.6	64.5	63.3
	10%	DB	66.4	65.8	68.4	67.7	70.0	73.2	77.2	75.7	74.9	72.3	70.1	66.9
		MCWB	60.6	60.2	61.6	63.1	65.6	66.8	67.3	68.1	67.2	66.8	63.9	62.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.9	66.5	68.0	68.0	70.0	73.9	74.0	73.6	73.5	71.2	70.1	67.2
		MCDB	69.3	69.7	73.0	70.5	75.2	85.2	90.8	91.1	81.6	75.2	74.3	69.3
	2%	WB	65.1	64.3	65.9	66.3	68.2	70.5	72.2	71.8	71.0	69.5	68.2	65.9
		MCDB	68.0	67.4	69.9	69.1	72.4	76.1	86.3	83.5	77.1	73.3	72.4	68.2
	5%	WB	63.0	62.7	64.2	65.0	66.8	68.9	70.2	70.2	69.5	68.3	66.6	64.7
		MCDB	66.3	65.6	68.0	67.9	70.9	73.8	78.8	78.4	73.6	72.2	70.5	67.1
	10%	WB	61.1	61.1	62.6	63.8	65.6	67.6	68.8	68.8	68.3	67.3	64.9	63.3
		MCDB	64.7	64.5	66.6	66.9	69.6	71.9	74.8	74.4	72.6	71.1	68.8	65.9
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	12.6	13.7	10.6	9.9	10.4	10.7	10.0	9.8	9.0	11.7	11.0
		MCDBR	19.3	17.5	20.1	17.1	17.9	20.5	25.4	23.3	20.3	15.9	18.1	13.0
		MCWBR	12.9	11.0	12.2	9.6	9.0	8.2	9.4	8.2	8.5	8.1	10.9	10.0
	5% WB	MCDBR	12.0	11.3	13.5	11.5	12.9	12.6	18.9	18.1	14.0	10.7	11.6	10.1
		MCWBR	10.5	9.2	10.8	8.2	8.2	7.5	8.2	7.7	7.8	7.2	9.1	8.7
(40) Clear-Sky Solar Irradiance	taub		0.329	0.364	0.416	0.413	0.434	0.469	0.531	0.526	0.448	0.412	0.370	0.345
	taud		2.509	2.398	2.213	2.259	2.217	2.123	1.940	1.977	2.212	2.313	2.417	2.475
	Ebn at Noon		285	283	274	279	272	262	245	245	261	262	266	272
	Edh at Noon		28	34	44	43	46	50	60	57	43	36	30	28
(44) All-Sky Solar Radiation	RadAvg		1118	1395	1794	2132	2358	2466	2465	2237	1903	1446	1149	1000
	RadStd		62	91	82	93	109	81	70	64	65	70	75	70

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.43	N/A	N/A	N/A	N/A	-0.58	N/A	N/A	N/A	+275	+308

Nomenclature: See separate page